SMART AYURVEDIC DISEASE PREDICTION AND RECOMMENDATION SYSTEM

Project ID: 25 - 26J – 305

Project Proposal Report

Fernando K.P.M.R.A – IT22897176

Supervised by Ms. Manori Gamage

Co-supervised by Ms. Uthpala Samarakoon

B.Sc. (Hons) Degree in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

August 2025

Source-Grounded Intelligent Question Answering (QA) over Classical Ayurvedic Literature

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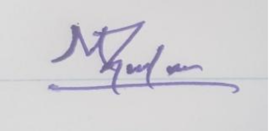
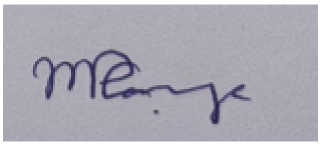
DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

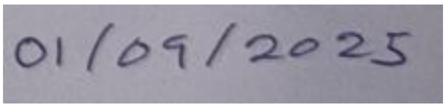
Name	Student ID	Signature
Fernando K.P.M.R. A	IT22897176	

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for their undergraduate dissertations under my supervision.



Signature of the supervisor:



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ABSTRACT

Classical Ayurvedic texts such as the Charaka Samhita, Madhava Nidana, and Ashtanga Hridaya represent more than two millennia of medical knowledge. Despite their historical and clinical value, these manuscripts remain computationally inaccessible due to multi-script formats, complex verse–commentary structures, and the absence of standardized digitized corpora. Modern Ayurvedic knowledge systems rely primarily on secondary explanatory materials rather than authoritative classical sources. As a result, practitioners, students, and researchers lack a mechanism to query and verify what original Ayurvedic texts actually state.

This research proposes a **Source-Grounded Question Answering System** capable of processing raw, multi-script classical Ayurvedic texts and returning accurate, verifiable answers supported by direct citations. The system introduces a unified framework that integrates **multi-script OCR**, **document structure recognition**, **cross-lingual semantic search**, and **source-grounded answer synthesis**. Users may pose questions in Sinhala, English, or Singlish, and the system retrieves relevant Sanskrit or Sinhala passages, synthesizes an answer, and provides exact citations for verification.

Novel contributions include:

- The first document intelligence pipeline optimized classical Ayurvedic manuscripts.
- A cross-lingual retrieval system that maps modern queries to Sanskrit medical terminology.
- A source-grounded QA model that fuses extractive and abstractive answer generation while ensuring citation faithfulness.
- A comprehensive evaluation framework for OCR accuracy, retrieval quality, and answer reliability.

This work addresses a critical gap in digital Ayurveda, enabling authentic knowledge access, supporting academic research, preserving cultural heritage, and providing a high-value standalone contribution to computational linguistics and traditional medicine informatics.

Keywords:

Ayurveda, Intelligent Question Answering, Semantic Search, Document Intelligence, OCR, Sanskrit NLP, Information Retrieval

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1 INTRODUCTION

One of the most ancient civilizations in the world boasting an extensive tradition of alternate medicine, the rich history of the ancient science of Ayurveda dates back thousands of years in the subcontinents of India as well as Sri Lanka. Inherently based on natural scientific systems as well as comprehensively grounded in philosophical expertise, the body of knowledge in the art of Ayurveda has been preserved in the form of the ancient treatises of Charaka Samhita, Ashtanga Hridaya, as well as the work of Madhava Nidana, along with the Sinhalese commentaries written by scholars over the centuries.

Although the importance of these ancient works has continued over the centuries, currently, they are unable to be accessed in the modern digital age. The manuscripts exist largely in printed form, if available at all, as PDFs, possibly degraded, written in the classic Sanskrit language, as well as in the classic Sinhalese form. The complexity of the manuscripts, ranging from poetry, commentary, to sub-commentary, along with the usage of older terms, defies efforts to understand the contents, particularly for students, practitioners, as well as researchers in Sri Lanka, where communication might be in the Sinhalese language, Code-mixed Sinhalese-English, also known as 'Singlish', while the digital information exists in the English language.

On the other hand, the past decade has seen significant advances in the area of Artificial Intelligence (AI), specifically in the fields of document intelligence, semantic search, as well as grounded question answering. Contemporary systems for question answering are capable of analyzing massive volumes of text, extracting accurate, domain-specific answers to natural language queries. These systems have been successfully applied to the fields of biomedical research, legal analytics, digital libraries, as well as customer support. The applications of these systems in the domain of Ayurveda, so far, remain sparse, piecemeal, as well as limited in functionality. Current systems in this domain are largely based on the rule-following approach that supports the English language alone or lack the functionality to link answers to the set of fundamental sutras, as well as commentaries that form the body of Ayurvedic knowledge. Specifically, there exists no system that can analyze the manuscripts directly, extracting text, analyzing structure, performing semantic search, as well as grounded answers based upon them.

In order to overcome this gap, this research proposes the Source-Grounded Question Answering System for Classical Ayurvedic Literature in an effort to span the gap between the ancient manuscripts and the modern digital provision. The innovation in this work comes from the fully integrated pipeline based on the following multi-level technologies:

- **Multi-script OCR** for text extraction from Sanskrit and Sinhala manuscripts.
- **Document intelligence** capabilities related to the recognition of verses, commentaries, and hierarchies.
- **Cross-lingual semantic search** supporting queries in the Sinhala, English, as well as Singlish.
- **Source-grounded QA** that generates answers strictly supported by classical textual evidence

Unlike existing chatbots or rule-driven Ayurvedic systems, the proposed framework does not generate unsupported interpretations or modern paraphrases. Instead, it provides answers backed by direct citations from classical texts-ensuring transparency, accuracy, and academic trustworthiness. This approach not only enhances user accessibility but also safeguards the integrity of the original Ayurvedic knowledge system.

In summary, this research bridges traditional Ayurvedic wisdom with modern AI methodologies, offering a transformative solution that is multilingual, source-grounded, and deeply aligned with the needs of both academic and practical communities. It introduces a robust digital infrastructure for processing classical texts and contributes novel computational resources for low-resource languages and culturally embedded medical knowledge.

1.1 Background & Literature survey

The Advancement of QA Systems

Ayurveda, one of the world's most influential traditional medical systems, represents more than two millennia of knowledge on human physiology, pathology, preventive care, and therapeutic practices. Foundational treatises such as the Charaka Samhita, Ashtanga Hridaya, and Madhava Nidana preserve highly structured medical insights written in classical Sanskrit verse, accompanied by Sinhala and Sanskrit commentaries authored by scholars across generations. These manuscripts form the authoritative basis of Ayurvedic education and clinical reasoning.

However, accessing these classical sources in the modern digital era remains extremely challenging. Most existing digital versions of Ayurvedic texts are limited to scanned PDFs or unstructured text, lacking semantic segmentation, searchable indexing, or machine-readable formatting. Linguistic barriers further complicate accessibility: Sri Lankan students, practitioners, and researchers commonly use Sinhala or code-mixed Sinhala–English (Singlish), while the majority of digitized Ayurvedic material exists in English or untranslated Sanskrit.

As a result, contemporary learners rarely interact with primary textual evidence but instead depend on secondary interpretations, which are prone to variation and inaccuracies. This gap limits academic inquiry, reduces clinical precision, and prevents the general population from engaging with authentic Ayurvedic knowledge.

In recent years, rapid advancements in Document Intelligence, Natural Language Processing (NLP), and Question Answering (QA) systems have transformed how humans interact with large bodies of text. Modern AI systems can extract information from complex documents, perform semantic retrieval across languages, and generate grounded answers supported by textual evidence. These technologies have reshaped domains such as biomedical literature mining, digital humanities, legal reasoning, and scientific research.

Despite this global progress, Ayurveda has not yet benefited from deep, structured, AI-driven digitization. Existing attempts are scattered, shallow, limited in linguistic scope, or lack transparency. Today, there is no system capable of performing multi-script OCR on classical Ayurvedic manuscripts, reconstructing hierarchical verse–commentary structures, and enabling cross-lingual semantic search with source-grounded question answering.

This research proposes a comprehensive solution to this gap.

Previous Ayurvedic QA Systems

- **AyurMate (2025) –** It is a rule-based assistant for matching symptoms with remedies. It performed well with established cases but was rigid when dealing with open-ended inquiries and did not support multiple languages [2].
- **VrikshBot (2025) –** This retrieval-based assistant is focused on the health of plants (Vrikshayurveda). It is limited only to plant care and operates solely in English [1].
- **LeNet + Stacked Autoencoders (2024) –** This deep learning model is utilized for categorizing Ayurvedic diseases. It is not a QA system and is confined to predictive functions [3].
- **AyUR-bot (2024) –** This hybrid RAG (retrieval + generation) system excels in English QA but falls short in terms of personalization, multilingual capabilities, and providing explanations [4].
- **AyurChat (2024) –** This chatbot employs NLP for intent detection and classification, offering conversational replies but lacking retrieval grounding or references to sutras [5].
- **Sinhala Ayurvedic Chatbot (2019) –** This system integrated CNN-based plant recognition with a Sinhala AIML chatbot. It was innovative at the time but limited by predefined rules and did not include retrieval or generative QA capabilities [6].

Previous systems share the following **limitations:**

- No multi-script OCR for Sanskrit + Sinhala manuscripts
- No structure-aware verse–commentary parsing
- No cross-lingual semantic retrieval (Sinhala/Singlish → Sanskrit)
- No source-grounded answers with verifiable citations
- No AI model trained on classical Ayurvedic literature
- No Sinhala or Singlish QA datasets for Ayurveda

These developments indicate gradual advancements but do not provide thorough, multilingual, source-grounded answers with citations & explainable QA system based on classical sources.

System	Dataset Size	QA Performance	Structure-Aware	Language Support	OCR Capability	Limitations/ Advantages
AyurMate (2025)	Small handcrafted symptom–remedy dataset (500 Q–A pairs)	N/A (rule-based)	No	English only	No	Rigid, Rule - based , Limited Scope
VrikshBot (2025)	Plant-health dataset (1,000 records)	80% (domain -specific classification)	No	English + Sanskrit	No	Limited to Plant health, not Medical QA
LeNet + SAE(2024)	2000 Ayurvedic Disease Cases	87 -90% (Classification Accuracy)	No	English only	No	Not QA, Classification only
AyUR-bot (2024)	10000 text passages indexed with FAISS	BLEU 65%, QA F1 72%	No	English only	No	No Personalization , No Multilingual support
AyurChat (2024)	800 intent and canned responses	75% (intent classification)	No	English only	No	Conversational but not Retrieval - based
Sinhala Ayurvedic Chatbot (2019)	Small Handcrafted Sinhala dataset (300 queries)	70% (pattern matching)	No	Sinhala only(No Cross-Lingual support)	No	Pattern-matching only (no semantic search), no source citations, no retrieval capability

Proposed System (2025)	Target 5000+ annotated QA pairs (Sinhala/English/ Singlish)	>85 (Target f1 for QA. Citation Accuracy >90%, Faithfulness Score >85%, BLEU >70%	Yes. Verse / Commentary (with Citations)	Sinhala / English / Singlish	Sanskrit Devnagari + Sinhala	First end-to-end classical Ayurveda QA

Table 1 Comparison of Existing Ayurvedic QA Systems

These systems highlight a gradual shift from rule-based Ayurvedic tools to more sophisticated retrieval and hybrid NLP techniques. However, they all share fundamental limitations: they do not process classical Ayurvedic manuscripts, they cannot handle Sinhala or code-mixed inputs, and no one provides verifiable, source-grounded answers from authoritative Ayurvedic texts. Their outputs are generally disconnected from the original Sanskrit sutras, limiting their academic value and undermining user trust.

Recent advancements in Question Answering (QA) technologies, especially in **document intelligence**, **semantic retrieval**, and **retrieval-augmented generation (RAG)** have opened significant new opportunities for bringing classical scholarly texts into digital, searchable, and explainable formats. Contemporary models such as LayoutLM, DocTr, mBERT, and large transformer-based QA frameworks allow systems to extract structured information from scanned manuscripts, understand layout and hierarchy, retrieve semantically related passages, and generate answers supported by textual evidence. Unlike previous rule-based or intent-classification chatbots that depend on predefined responses, source-grounded QA frameworks enable systems to consult primary literature directly. This offers the potential to return accurate, contextually meaningful, and citation-backed answers drawn from original Ayurvedic treatises such as the Charaka Samhita, Ashtanga Hridaya, and Madhava Nidana.

Another major advancement relevant to this research is the emergence of **multi-script OCR and post-correction pipelines**, which can be adapted to read degraded Sanskrit and Sinhala manuscripts. Existing tools-while not optimized for Ayurveda, demonstrate that combining OCR with domain-specific correction dictionaries significantly improves accuracy. For Ayurvedic manuscripts, which contain non-standard ligatures, archaic fonts, and mixed Sanskrit–Sinhala layouts, specialized OCR workflows are essential and currently absent from any existing system.

Equally significant is the development of **cross-lingual semantic search**, which enables queries in Sinhala, English, or Singlish to be mapped to Sanskrit and classical Sinhala content. Prior Ayurvedic systems like AyurChat or AyUR-bot function exclusively in English and rely on simple keyword matching. They are unable to understand Sinhala terms, transliterated expressions, or colloquial phrasing commonly used by students and practitioners in Sri Lanka. By incorporating multilingual embeddings, Sinhala–Sanskrit alignment models, and transliteration handling, the proposed system addresses a linguistic gap that has never been explored in previous Ayurvedic research.

Furthermore, the new system introduces **structure-aware document intelligence**, enabling it to distinguish sutras, commentaries, sub-commentaries, and chapter metadata—an essential feature for any system interacting with classical Ayurvedic literature. Previous systems treat texts as flat strings without understanding the canonical hierarchy, which prevents accurate retrieval and breaks contextual relationships between verses and their explanations.

Finally, the proposed framework ensures **sutra-linked explainability**, providing answers that are directly tied to specific verses and commentary passages. This stands in sharp contrast to existing systems that generate responses without citations, risking misinformation and reducing scholarly reliability. By grounding every answer in primary Ayurvedic sources, the system enhances transparency, academic rigor, and user confidence.

Together, these innovations multi-script OCR, document intelligence, cross-lingual retrieval, and source-grounded QA demonstrate how the proposed system addresses long-standing gaps in digitizing and accessing classical Ayurvedic knowledge. In contrast to earlier technologies that were limited, monolingual, or disconnected from classical texts, this research aims to develop the first end-to-end system that transforms handwritten and printed Ayurvedic manuscripts into a structured, searchable, and reliable knowledge base capable of delivering multilingual, evidence-linked answers.

1.2 Research Gap

Classical Ayurvedic knowledge is preserved in multi-script Sanskrit and Sinhala manuscripts that contain thousands of verses, commentaries, and highly specialized medical terminology. Although digitization efforts exist in scattered forms, no system currently provides structured, searchable, or interpretable access to these texts. The literature identifies several isolated advancements -Indic OCR models, Sanskrit digital libraries, biomedical QA systems, and multilingual embeddings. However, these efforts remain fragmented and do not integrate into a unified, practical solution for Ayurvedic knowledge retrieval.

Based on the literature review, the following major research gaps are identified:

1.2.1 Absence of a Domain-Adapted OCR Pipeline for Classical Ayurveda

Existing OCR engines are inadequate for classical Sanskrit and Sinhala manuscripts:

- They cannot accurately recognize ligature-heavy Sanskrit characters
- They fail to differentiate verse, commentary, headers, and sub-commentary.
- They do not handle traditional Sinhala fonts present in Ayurvedic print editions.
- No OCR model incorporates Ayurvedic medical dictionaries for post-correction.

There is no standardized pipeline that converts raw manuscript/PDF pages into structured digital text that preserves the logical order of the original work.

1.2.2 Lack of Document Structure Recognition for Ayurveda Texts

Modern Document AI tools (DocTr, LayoutLM) do not support:

- Verse boundary identification.
- Layered commentary segmentation.
- Sanskrit → Sinhala cross-notes.

- Identification of chapter sections within ancient books.
- Linking commentary back to the originating verse.

This gap prevents meaningful computational analysis of classical texts.

1.2.3 No Cross-Lingual Semantic Search System Built for Sanskrit Medical Terminology

Contemporary dense retrieval models:

- Ignore Sanskrit morphology.
- Cannot process Sinhala queries.
- Do not embed classical medical vocabulary.
- Cannot map Singlish queries (e.g., "rasa balancing foods") to Sanskrit verses

This is a major barrier for modern users accessing historical knowledge.

1.2.4 No Source-Grounded QA System for Classical Ayurveda

Existing QA systems either:

- Produce hallucinations.
- Lack citation faithfulness.
- Operate only in modern languages.
- Require structured datasets, which do not exist for Ayurveda.
- Do not support long-document attention required for verse + commentary retrieval

The inability to return:

- Exact classical citations.
- Combined verse + commentary meaning,
- Multi-script outputs

is a gap of high academic and practical importance.

1.2.5 Lack of a Unified End-to-End Framework

Even though individual components exist in other domains (OCR, embeddings, QA), **there is no integrated framework** for:

1. Document ingestion.

2. Text extraction and structure reconstruction.
3. Cross-lingual semantic retrieval.
4. Source-grounded answer generation.
5. Citation verification

Such a pipeline is essential but currently non-existent.

1.3 Research Problem

Based on the identified research gaps, the overarching problem addressed by this study can be formulated as follows:

"How can a unified system be developed to extract, structure, search, and generate answers from classical Ayurvedic manuscripts in a way that supports natural-language queries while ensuring verifiable citations from authoritative sources?"

This central problem can be decomposed into four critical sub-problems:

Problem 1: Extracting Reliable Text from Multi-Script Classical Manuscripts

Classical Ayurvedic manuscripts exist in varied formats, including:

- Devanagari Sanskrit text
- Sinhala commentary
- mixed-layout PDF scans
- degraded print editions

Challenges include:

- low OCR accuracy
- font variations
- verse–commentary interleaving
- absent annotations

The system must accurately convert these diverse inputs into structured digital text.

Problem 2: Retrieving Semantically Relevant Passages across Languages

Users ask questions in Sinhala, English, or Singlish.

The system must:

- Map modern vocabulary to Sanskrit medical terms
- Understand conceptual similarity

- Retrieve correct verses and commentaries
- Filter noisy or unrelated matches

This requires a cross-lingual embedding model specific to classical Ayurvedic content.

Problem 3: Providing Faithful, Source-Grounded Answers with Citations

A valid answer must include:

- synthesized explanation
- original Sanskrit or Sinhala passage
- citation details (chapter, verse number)

  Current QA models hallucinate or paraphrase inaccurately.

  The challenge is designing a QA module that:
- grounds answer strictly in retrieved text
- verifies citations
- generates explanations without altering classical meaning

This is the most critical part of the research.

Problem 4: Understanding the Hierarchical Structure of Ayurvedic Texts

Ayurvedic texts follow a unique hierarchical structure:

```
|- Chapter
    |-- Section
      |----Verse
         |------Commentary
            |-------- Sub-commentary
```

The problem is:

- Standard document extraction flattens this hierarchy
- Computational models fail to detect Sanskrit verse boundaries
- Commentary cannot be correctly linked back to verses

Without structure reconstruction, semantic retrieval becomes unreliable.

Consequently, the primary research question can be articulated as:

“How can a multilingual, constitution-aware, and explainable QA system for Ayurveda be created that incorporates retrieval-augmented generation, domain-specific named entity recognition, and sutra-linked citations, given the lack of structured datasets?”

This question encompasses four interconnected challenges:

- Overcoming language barriers by accommodating multilingual and code-mixed inquiries.
- Facilitating personalization by embedding prakriti and dosha profiles into the QA framework.
- Promoting explainability by anchoring responses in recognized sutras.
- Mitigating data scarcity by developing a structured Ayurvedic QA dataset.

Tackling these challenges necessitates a shift from rule-based or retrieval-only models towards a hybrid architecture that combines retrieval methods, generative transformers, named entity recognition, and dedicated personalization layers, ultimately linking the ancient wisdom of Ayurveda with contemporary AI-powered question-answering systems.

2 OBJECTIVES

The development of a Source-Grounded Question Answering (QA) system for classical Ayurvedic manuscripts requires a set of objectives that integrate modern AI methodologies with the unique textual, linguistic, and structural characteristics of traditional Ayurvedic literature. Unlike biomedical QA systems, which benefit from abundant structured datasets and standardized terminology, the Ayurvedic domain poses distinct challenges: multi-script manuscripts (Sanskrit and Sinhala), degraded historical scans, the absence of machine-readable corpora, and the need for meaningful retrieval without relying on translation. This research therefore aims to combine **multi-script OCR**, **document intelligence**, **cross-lingual semantic retrieval**, and **source-grounded answer generation** into a single, end-to-end framework capable of making classical manuscripts accessible, searchable, and explainable to modern users.

The primary aim is to enhance access to authentic Ayurvedic knowledge by enabling users to query ancient manuscripts in Sinhala, English, or Singlish and receive answers supported directly by verses from authoritative texts such as Charaka Samhita, Ashtanga Hridaya, and Madhava Nidana. In contrast to earlier Ayurvedic chatbots which rely on predefined rules, English-only interfaces, or generalized paraphrasing the proposed system emphasizes faithfulness, transparency, and scholarly accuracy. By grounding responses in classical sutras and reconstructing manuscript structure through advanced document AI, this research bridges modern computational methods with the depth and integrity of traditional medical literature.

2.1 Main objective

The main objective of this research is:

To design and develop an end-to-end, source-grounded Question Answering system that digitizes classical Ayurvedic manuscripts using multi-script OCR, reconstructs their verse–commentary structure with document intelligence, enables cross-lingual semantic retrieval for Sinhala, English, and Singlish queries, and generates citation-backed answers grounded strictly in authoritative Ayurvedic texts.

This objective focuses on **accessibility**, **authenticity**, and **explainability**, ensuring that users can interact with classical sources directly and receive reliable answers supported by meaningful evidence.

2.2 Specific objective

To achieve the main objective, this research defines the following Specific Objectives, aligned with SMART principles:

- **Develop a multi-script OCR workflow capable of accurately extracting Sanskrit (Devanagari) and classical Sinhala text from scanned Ayurvedic manuscripts.**

  This includes preprocessing of degraded pages, script segmentation, OCR fine-tuning, and domain-specific post-correction using Ayurvedic lexicons to ensure high-quality text digitization.

- **Implement a structure-aware document intelligence pipeline to automatically identify and classify verses, commentaries, sub-commentaries, section headers, and reference markers in classical texts.**

  This ensures hierarchical reconstruction of manuscripts, allowing accurate retrieval and contextual understanding.

- **Create a cross-lingual semantic retrieval engine that maps user queries in Sinhala, English, or Singlish to Sanskrit or Sinhala manuscript content without relying on direct translation.**

  The system will use multilingual embeddings and hybrid retrieval (dense + sparse) to locate relevant sutras based on semantic meaning.

- **Develop a source-grounded QA module that generates human-readable explanations in Sinhala or English while strictly grounding statements in retrieved classical passages.**

  This component must ensure citation accuracy, prevent hallucinations, and preserve the original meaning of the sutras.

- **Build the first machine-readable corpus of structured Ayurvedic manuscript data, including verse segmentation, commentary alignment, metadata extraction, and semantic annotations.**

This corpus will serve as the foundational dataset for retrieval, QA evaluation, and future computational Ayurveda research.

- **Evaluate the complete system using OCR accuracy metrics (CER/WER), retrieval metrics (Recall@K, MRR), QA metrics (F1, citation accuracy), and user-level feedback from Ayurvedic scholars and students.**

  The system aims to achieve high performance across OCR, retrieval, and QA, demonstrating improvements over existing baselines that lack manuscript grounding.

3 METHODOLOGY

3.1 System Architecture Overview

The Source-Grounded Question Answering System for Classical Ayurvedic Literature employs a comprehensive two-part architecture designed to transform computationally inaccessible classical manuscripts into a query able knowledge source while maintaining authenticity to original texts. Unlike conventional medical QA systems that rely on structured databases or secondary interpretations, this system processes raw classical manuscripts and provides intelligent, source-verified answers with direct citations to primary sources.
The architecture comprises two distinct operational phases:

Part A: Offline Document Intelligence Pipeline
This phase transforms raw classical manuscripts into a structured, searchable knowledge base through specialized document processing techniques. The offline pipeline consists of:

- **Document Processing Layer** - Responsible for multi-script OCR (Sanskrit Devanagari, Sinhala, Roman), layout analysis to distinguish verses from commentaries, and quality assessment of extracted text.
- **Text Intelligence Layer** - Performs Sanskrit normalization, sentence segmentation, medical entity tagging, and cross-reference linking to preserve the hierarchical structure of classical texts.
- **Indexing and Embedding Layer** - Creates both dense embeddings using domain-adapted multilingual transformers (mBERT, IndicBERT) and sparse representations using BM25/TF-IDF for hybrid retrieval capabilities.
- **Knowledge Base Creation** - Generates a comprehensive corpus of 2000+ processed pages from Charaka Samhita, Madhava Nidana, and Ashtanga Hridaya with metadata, structural preservation, and citation indices.

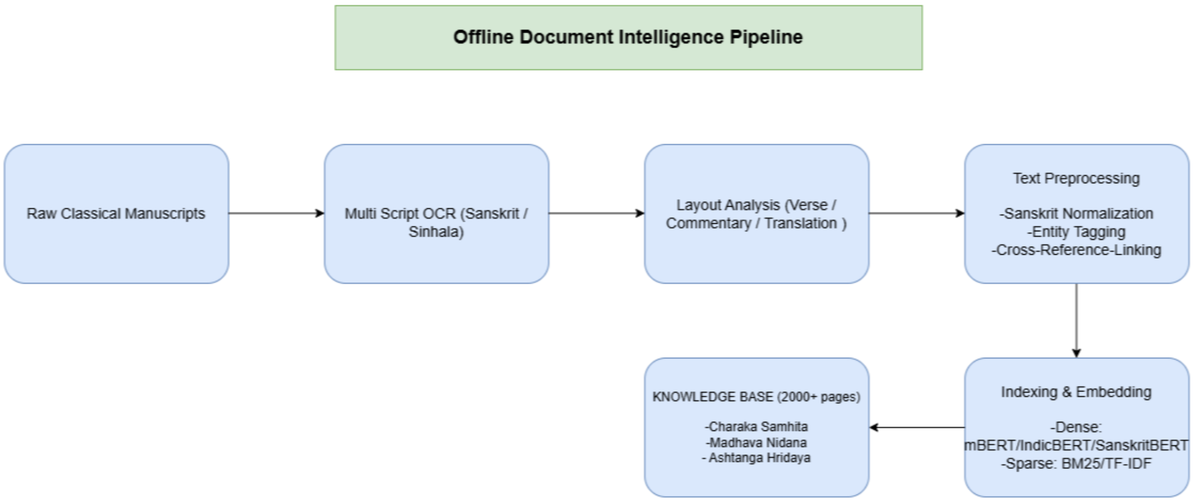

Figure 1 Offline Document Intelligence Pipeline

Part B: Online Query processing Pipeline

This phase handles real-time user queries and generates source-grounded answers through intelligent retrieval and synthesis. The online pipeline consists of:

- **Query Processing Layer** - Handles multilingual input (Singlish, English, Sinhala) through language detection, code-switching normalization, and medical term expansion with Sanskrit equivalent mapping.
- **Cross-Lingual Embedding Layer** - Transforms user queries into semantic representations using fine-tuned multilingual models, enabling cross-lingual retrieval from Sanskrit texts.
- **Semantic Search Layer** - Employs hybrid retrieval combining dense vector search (FAISS) and sparse keyword matching (BM25) with medical relevance re-ranking to retrieve the most pertinent passages.
- **Answer Generation Layer** - Synthesizes coherent answers from multiple retrieved passages while maintaining faithfulness to source texts, linking each claim to specific chapter and verse citations.
- **Explainability Layer** - Formats responses with Sanskrit citations, confidence scores, and source passage displays to enable verification and transparency. The structure of this architecture guarantees that responses are not only contextually precise and multilingual but also verifiable through direct source attribution, addressing the critical gap in accessing primary classical medical literature.

This represents the first computational framework specifically designed for processing and querying classical Ayurvedic manuscripts at scale.

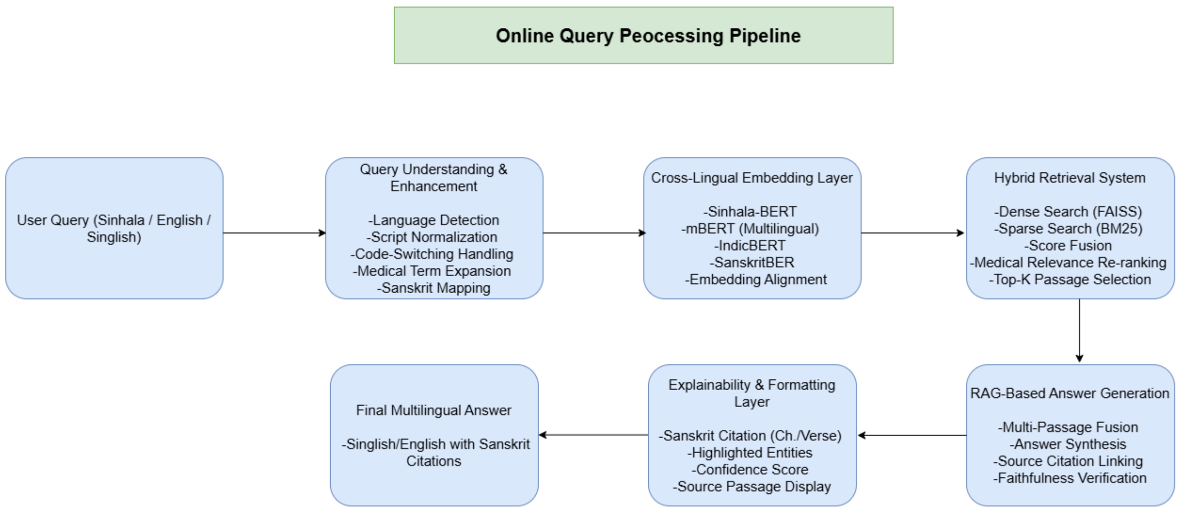

Figure 2 Online Query processing Pipeline

3.2 Data Collection and Corpus Creation

The success of a source-grounded QA system is fundamentally dependent on the quality and authenticity of its knowledge base. Unlike existing systems that rely on curated structured datasets, this research creates its corpus from raw classical manuscripts through specialized document processing. Data will be collected and processed from three categories of sources:

- **Classical Ayurvedic Manuscripts** - Primary focus on digitized versions (both printed editions and manuscript scans) of Charaka Samhita (targeting 800 pages across 8 sections), Madhava Nidana (targeting 600 pages focusing on disease descriptions across 78 chapters), and Ashtanga Hridaya (targeting 400 pages from the comprehensive compendium). These texts will be processed in their original Sanskrit Devanagari and Sinhala scripts.
- **Parallel Translations and Commentaries** - Sinhala translations and traditional commentaries (targeting 200 pages) will be collected to enable cross-lingual understanding and provide context for Sanskrit verses. These parallel texts are essential for training cross-lingual embeddings and validating translation accuracy.
- **Validation and Benchmark Datasets** - To evaluate system performance, structured QA datasets will be created with assistance from Ayurvedic scholars and practitioners.

  This includes:

  - Document Processing Validation Set: 200+ manually annotated pages for OCR accuracy assessment.

  - Retrieval Benchmark Set: 300+ query-document pairs with relevance judgments.

- QA Evaluation Set: 500+ question-answer pairs with source citations for end-to-end evaluation.
- Medical Entity Dataset: 3000+ annotated sentences for NER training covering diseases, herbs, doshas, symptoms, and treatments.

The corpus creation process involves extensive manual validation to ensure OCR accuracy, proper structural preservation, and correct entity annotations. Inter-annotator agreement (Cohen's Kappa >0.85) will be maintained for all annotated datasets through collaboration with Ayurvedic experts and computational linguists. The goal is to create the first comprehensive, computationally accessible corpus of classical Ayurvedic texts with preserved structure, accurate transcription, and rich metadata.

3.3 Document Intelligence and Corpus Processing

Processing classical Ayurvedic manuscripts presents unique computational challenges distinct from modern document processing. These texts feature multiple scripts, complex hierarchical layouts (verses, commentaries, sub-commentaries), specialized medical terminology, and ancient typesetting conventions. To address these challenges, a specialized document intelligence pipeline will be implemented with the following components:

1. **Multi-Script OCR and Image Preprocessing**

   All manuscript images will undergo preprocessing including noise reduction, binarization, and deskewing to optimize OCR performance. A hybrid OCR approach combining Tesseract OCR and PaddleOCR will be employed, with custom training data specifically designed for:

   - Sanskrit Devanagari medical texts (Charaka Samhita, Madhava Nidana)
   - Sinhala script (translations and commentaries)
   - Roman transliteration (modern editions).

   The OCR pipeline will include quality assessment algorithms that compute character-level and word-level confidence scores, flagging low-quality regions for manual review. Target OCR accuracy: >90% for printed texts, >80% for manuscript scans.

2. **Layout Analysis and Structure Recognition**

   Classical texts have distinctive structural patterns that must be preserved for accurate information retrieval. A layout classification module will be trained to automatically identify:

   - Main verses (Sutras) – primary canonical statements
   - Primary commentaries (Bhasya) – traditional interpretations
   - Sub-commentaries (Tika) – additional explanatory layers
   - Translation sections – modern language equivalents

This structure-aware processing ensures that answers can cite specific verse numbers and distinguish between original text and interpretative commentary. Target layout classification accuracy: >85% F1-score.

3. **Text Standardization and Preprocessing**

Extracted text undergoes several normalization steps:

-Sanskrit Normalization: Standardization of variant spellings, sandhi resolution, and transliteration to IAST (International Alphabet of Sanskrit Transliteration)

- Sinhala Normalization: Unicode standardization and variant character resolution
- Sentence Segmentation: Verse boundary detection and sentence-level chunking
- Error Correction: Domain-knowledge-based correction of OCR errors using medical terminology dictionaries

This ensures computational uniformity across texts while preserving semantic integrity

4. **Medical Named Entity Recognition (NER)**

A specialized NER model will be developed to identify and tag Ayurvedic entities within the processed text:

- Diseases and Conditions (Rogas): e.g., Jwara (fever), Atisara (diarrhea) - Target: 500+ entities.
- Medicinal Herbs (Aushadhis): e.g., Ashwagandha, Triphala, Haritaki - Target: 800+ entities.
- Constitutional Elements (Doshas): Vata, Pitta, Kapha and their combinations - Target: 300+ entities
- Symptoms (Lakshanas): Clinical signs and manifestations - Target: 600+ entities.
- Treatments (Chikitsa): Therapeutic procedures and formulations - Target: 400+ entities
- Textual References: Cross-references to other chapters and texts

NER training will utilize a manually annotated dataset of approximately 3,000 sentences with contributions from Ayurvedic experts. The model will be fine-tuned on IndicBERT/mBERT architectures with target performance >80% F1-score across all entity types.

5. **Cross-Reference Linking and Metadata Generation**

The system will automatically detect cross-references within texts (e.g., "as mentioned in Sutra Sthana, Chapter 25") and create bidirectional links. Metadata including chapter numbers, verse numbers, section titles, and thematic tags will be extracted and associated with each text segment to enable precise citation.

This comprehensive document processing pipeline represents a novel contribution to digital humanities, being the first framework specifically designed for classical medical Sanskrit literature. The processed corpus becomes the foundation for all downstream retrieval and answer generation tasks.

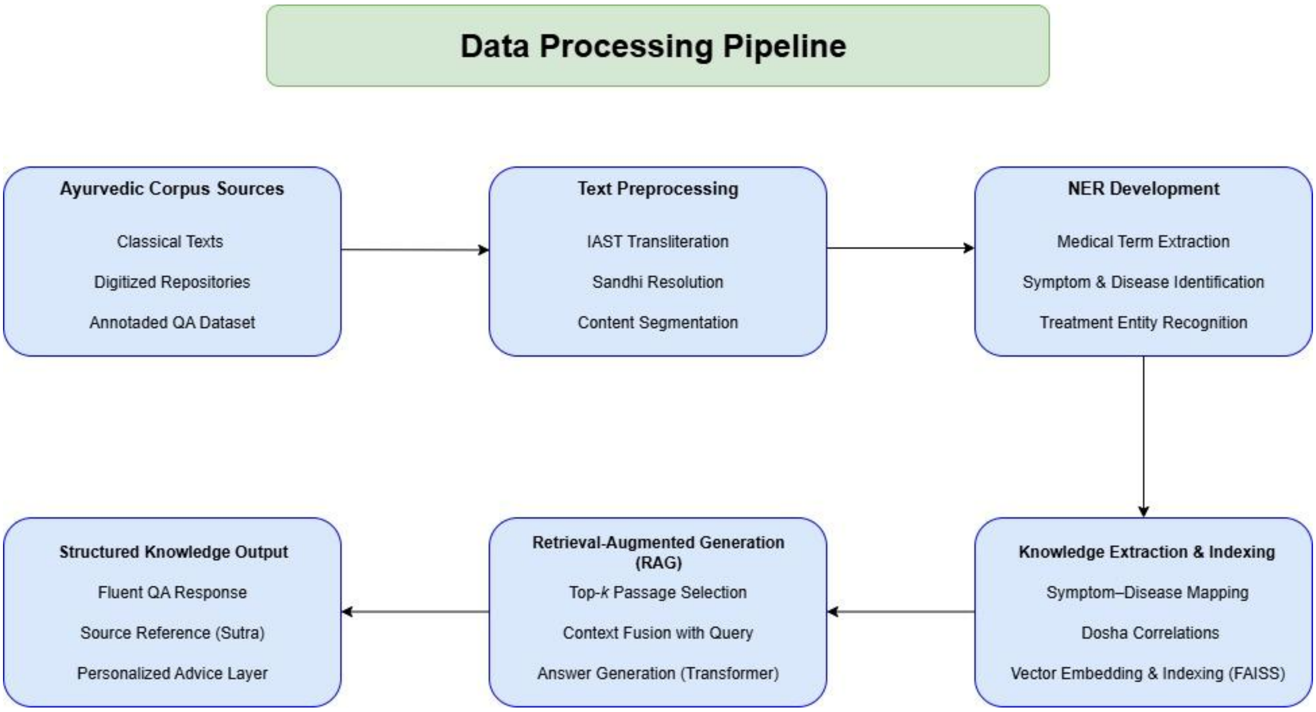

Figure 3 Data Processing Pipeline of the Ayurvedic QA Module

3.4 Cross-Lingual Semantic Search and Retrieval

The retrieval component must address a fundamental challenge: enabling users to query in colloquial languages (Singlish, English, Sinhala) while searching through Sanskrit classical texts. This requires sophisticated cross-lingual semantic understanding and hybrid retrieval mechanisms that balance semantic similarity with terminological precision.

Query Understanding and Enhancement

Before retrieval, user queries undergo multi-stage processing:

1. **Language Detection and Normalization:** Automatic detection of input language (Sinhala, English, Singlish, or code-mixed), followed by script normalization and transliteration handling.
2. **Code-Switching Resolution:** Singlish queries (e.g., "mata thahanavak, isa vedanawa athi, ayurveda kiyanne mokakda?") are analyzed to identify language boundaries and normalize mixed expressions.
3. **Medical Term Expansion:** User queries are enhanced with medical synonyms and related terms. For example, "fever" is expanded to include "jwara", "santapa", "ushnata".

4. **Sanskrit Equivalent Mapping:** Modern medical terms are mapped to their classical Sanskrit equivalents using a curated terminology database. This enables semantic matching between colloquial queries and formal classical descriptions.

Hybrid Retrieval Architecture

The system employs a hybrid retrieval approach combining the strengths of dense semantic search and sparse keyword matching:

Dense Retrieval:

- Query Encoding: User queries are encoded using fine-tuned cross-lingual embeddings (mBERT, IndicBERT, Sinhala-BERT).
- Document Encoding: All text segments in the corpus are pre-encoded and indexed in FAISS vector database.
- Semantic Search: Top-k passages (k=20) are retrieved based on cosine similarity in the shared embedding space.

This captures semantic relatedness even when exact terms don't match.

Sparse Retrieval:

- Inverted Index: Classical BM25 index created using Elasticsearch for exact term matching.
- Keyword Extraction: Important medical terms from queries are identified and searched.
- Passage Ranking: Top-k passages retrieved based on term frequency and inverse document frequency.

This ensures passages with specific medical terminology are not missed.

Score Fusion and Re-ranking:

- Initial retrieval scores from dense and sparse methods are normalized and combined using weighted fusion.
- A medical relevance re-ranker (fine-tuned BERT-based cross-encoder) re-scores the top candidates considering both query-passage relevance and medical context appropriateness.
- Final top 5 passages are selected for answer generation.

This hybrid approach achieves the best of both worlds: semantic understanding from dense retrieval and terminological precision from sparse retrieval. Target performance: Precision@5 >80%, Recall@10 >70%, Mean Reciprocal Rank (MRR) >0.75.

Approach	Expected Accuracy	Training Time	Source Grounding & Citations	Advantages	Limitations
Retrieval-Based QA	70 – 75% (f1)	Fast (milliseconds)	Limited (passage retrieval only, no structured citations)	Reliable grounding in corpus, easy to implement, Fast response time	Answers are extractive, not conversational, struggles with multilingual & code-mixed queries.
Generative QA	75 – 80% (BLEU)	Moderate (seconds per query)	None (no source verification, no citations)	Produce fluent ,natural responses, good readability	May imagine responses, lacks sutra connection, weak factual basis, Lacks source citations.
Hybrid RAG QA (proposed)	85%+(F1) + BLEU >70%	Moderate (Retrieval + Generation)	Yes (Chapter / Verse citations, faithfulness verifications, source passage display)	Combines factual groundings with fluent generation, supports multilingual queries, sutra-linked explainability	Requires more computational resources, Pipeline complexity is higher.

Table 2 QA Model Comparison

Retrieval-Based QA

This baseline method relies exclusively on semantic search to extract passages from the corpus. Queries are transformed into embeddings using Sentence Transformers, and top-k passages are retrieved through FAISS vector search. While this ensures factual accuracy by grounding answers in the corpus, outputs are extractive snippets rather than natural language replies. Critically, it does not provide structured chapter and verse citations, making verification difficult.

Generative QA

Generative QA utilizes transformer models (mT5, IndicBERT) to generate fluent answers directly from queries without explicit retrieval. This provides conversational responses but suffers from hallucination-faithfulness scores of only 40-60%, which means nearly half the content is unsupported by actual texts. Without citations, verification is impossible.

Hybrid Retrieval-Augmented Generation (RAG) - Proposed Approach

The proposed system combines retrieval and generation advantages while addressing their weaknesses. Hybrid retrieval guarantees factual accuracy, while the generation module formulates fluent, multilingual answers strictly supported by passages. Faithfulness verification ensures >85% of content is source grounded. Structured chapter and verse citations enable verification against primary texts. Evaluation focuses on QA F1-score (>85%), BLEU (>70%), and two novel metrics: Citation Accuracy (>90%) and Faithfulness Score (>85%).

Overcoming Limitations: The system employs lightweight models (IndicBERT, distilled mT5) to reduce resource needs while preserving accuracy. FAISS is enhanced with quantization to lower memory usage. Caching retains common queries to reduce repeated calculations. Modular architecture segmenting into independent layers (Document Processing, Retrieval, Generation, Explainability) simplifies maintenance. Transfer learning from pre-trained models avoids training from scratch. Pipeline orchestration tools (MLFlow) optimize integration and monitoring.

3.5 Explainability and Response Formatting

The system emphasizes explainability through source-grounded transparency. Every response provides multiple layers enabling users to trace claims back to specific classical texts.

Source Attribution

Primary Citations: Answers include structured citations formatted as "Text Name, Section, Chapter Number, Verse Range" (e.g., "Charaka Samhita, Chikitsa Sthana, Chapter 3, verses 12-15").

Original Text Display: Sanskrit passages shown with Devanagari script, IAST transliteration, English translation, and Sinhala translation (if available).

Multiple Source Indication: When multiple texts confirm information, all sources are indicated (e.g., "mentioned in both CS.Chi.3.12-15 and AH.Chi.1.20-22"). Divergent sources are explicitly noted.

Entity Highlighting

Medical entities (diseases, herbs, doshas, treatments) are visually highlighted using color coding. An interactive glossary provides definitions on hover, with detailed explanations on click. Each answer suggests related entities and concepts for deeper exploration.

Confidence and Relevance Indicators

Overall Confidence Score: Displayed as numerical value (0-1) with interpretation (High/Medium/Low) and brief explanation.

Per-Statement Confidence: Individual confidence for each claim when answers contain multiple statements.

Relevance Score: Indicates how well retrieved passages match the query.

Source Quality Indicators: Distinguishes primary text from commentary, indicates multiple text agreement (e.g., "Confirmed across 3 classical texts").

Context and Additional Information

Related Passages: Suggestions for related information from other chapters with cross-references.

Modern Context: Careful connections to contemporary understanding when relevant, clarifying that classical and modern systems use different frameworks.

Related Questions: System suggests related queries users might be interested in.

Cautionary Notes: Explicit disclaimers for complex medical topics requiring professional consultation, with safety warnings for potentially harmful substances.

Response Format Template:

Question: [User's query in their language]

Answer: [Main answer text - 2-4 paragraphs with highlighted key terms]

Source Citations: 1. Charaka Samhita, Chikitsa Sthana, Chapter 3, verses 12-14 Original (Sanskrit): [Sanskrit text] Transliteration: [IAST] Translation: [English/Sinhala]

Confidence & Relevance:

- Overall Confidence: 0.87 (High)
- Source Agreement: 3 texts confirm
- Relevance Score: 0.91

Key Entities Mentioned: [Disease/Herb/Dosha] - [Category]

Related Information: [Cross-references to related chapters]

3.6 Evaluation Strategy

To assess the Source-Grounded QA system, a multi-faceted evaluation framework combining quantitative metrics, user studies, and expert validation will be employed.

Quantitative Evaluation Metrics:

1. Document Processing Metrics:
    - OCR Accuracy: CER <10% (printed), <20% (manuscripts)
    - Layout Classification: F1 >85%
    - NER Performance: F1 >80%
    - Processing Throughput: 5-10 pages/hour 2.
2. Retrieval Quality Metrics:
    - Precision@5: >80%, Precision@10: >75% .
    - Recall@5: >65%, Recall@10: >70%
    - Mean Reciprocal Rank (MRR): >0.75
    - NDCG@10: >0.80
3. Answer Generation Metrics:
    - QA F1 Score: >85%
    - BLEU Score: >70 (if abstractive)
    - BERTScore: >0.85
    - Answer Relevance: >4/5 (human-judged)
4. Source Grounding Metrics:
    - Citation Accuracy: >90% (expert validation) \
    - Faithfulness Score: >85% (NLI-based + human evaluation)
    - Hallucination Rate: <10%
    - Source Coverage: >95%

Approach	QA F1(Accuracy)	BLEU (Fluency)	Recall@k (Retrieval quality)	Citations Accuracy	Target Benchmark
Retrieval-Based QA	70 – 75%	N/A (Extractive only)	85%	N/A (No Citations)	Baseline factual grounding
Generative QA	75 – 80%	60 – 65%	N/A	N/A (No Citations)	Fluent but less reliable
Hybrid RAG QA (proposed)	> 85%	> 70%	> 90%	>85%	Balanced accuracy + fluency

Table 3 QA Evaluation Metrics Comparison

4 Component Diagram

The Intelligent Ayurvedic QA aims to create a trustworthy, easy-to-understand, and personalized question-answering system rooted in traditional Ayurvedic wisdom. It caters to user queries in Sinhala, English, and even a mix of both. By combining structured knowledge from respected Ayurvedic texts with cutting-edge NLP techniques, this component guarantees both accuracy and user-friendliness.

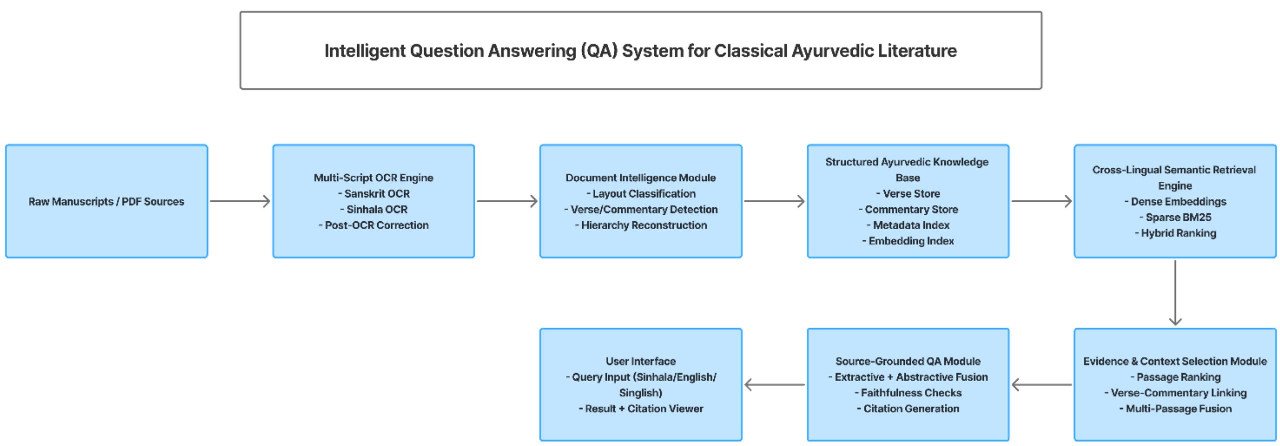

Figure 4 Component Diagram

5 Requirement Gatherings

The Ayurvedic QA system aims to support patients, students, and practitioners by delivering trustworthy, verifiable, and culturally appropriate responses grounded in classical texts. Unlike standard search engines or generic chatbots, this system provides precise chapters and verse citations enabling users to verify every claim against primary Ayurvedic sources.

5.1 Data Sources

To create a comprehensive knowledge base, data will be gathered from:

- Classical Ayurvedic manuscripts and texts (Charaka Samhita, Sushruta Samhita, Ashtanga Hridaya, Madhava Nidana) in original Sanskrit with commentaries.
- Multi-script digitized texts include Devanagari, Sinhala translations, and IAST transliteration.
- Digital archives and scholarly editions from Ayurvedic research institutions.
- Traditional Knowledge Digital Library (TKDL) for authenticated classical references.
- Ayurvedic healthcare facilities in Sri Lanka (Rajagiriya, Nawinna) for validation support.

This approach ensures authentic classical knowledge with proper source attribution for scholarly and clinical reliability.

5.2 Preprocessing Pipeline

The gathered data will undergo structured document intelligence processing:

- Multi-script OCR – extracting text from printed books and manuscript scans (Sanskrit Devanagari, Sinhala).
- Layout analysis – identifying verses, commentaries, chapter headers, and translations.
- Text normalization – standardizing transliterations, correcting OCR errors, verse segmentation.
- Structure preservation – maintaining chapter/verse numbering and cross-references.
- Entity annotation – labeling diseases, herbs, doshas, treatments with canonical forms.
- Citation indexing – creating searchable index mapping content to chapter/verse locations.
- De-duplication – eliminating repetitive passages across editions

This pipeline ensures a clean, semantically rich, and source-verifiable dataset enabling the QA system to provide precise, trustworthy responses with accurate citations.

6 Project Timeline

The research initiative will take place over a one-year duration, organized into distinct phases to guarantee methodical development and assessment. The schedule aims to harmonize ambitious research goals with practical implementation targets, considering resource availability, technical intricacies, and the necessity for iterative improvements. Each phase progressively relies on the preceding one, incorporating regular evaluations and expert validations throughout. The Gantt chart below illustrates the anticipated activities, dependencies, and expected outcomes across the project's timeline.

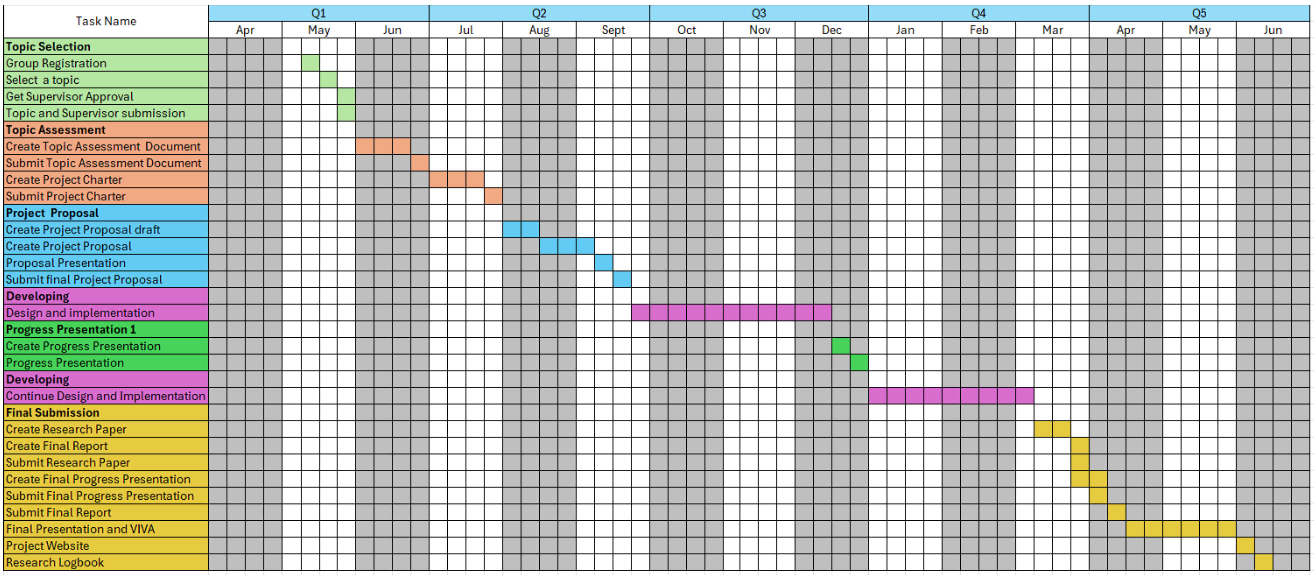

Figure 5 Gantt chart: The timeline for completing the development of the proposed system

7 PROJECT REQUIREMENTS

7.1 Functional requirements

7.1.1 User Authentication and Management

- The system shall facilitate secure registration and login for patients, students, and practitioners.
- Implement role-based access: general users (make inquiries), practitioners (provide validation feedback) .
- Offer password reset capabilities through secure email verification.
- The system shall maintain user profiles including query history and saved responses

7.1.2 Query Processing Module

- The system shall accept queries in multiple languages (Sinhala, English, mixed Sinhala-English).
- Standard queries addressing transliteration, synonyms, and spelling variations.
- Conduct tokenization and lemmatization for semantic consistency.
- The system shall detect code-switching in Singlish queries and resolve language boundaries.
- Map modern medical terms to classical Sanskrit equivalents for accurate retrieval

7.1.3 Named Entity Recognition (NER)

- The system shall identify Ayurvedic entities: doshas, diseases (roga), herbs (aushadha), treatments (chikitsa), body constituents (dhatu) .
- Facilitate differentiation of polysemous terms (e.g., "Vata" as dosha versus wind).
- Tag recognized entities with canonical Sanskrit names and translations .
- The system shall enable entity highlighting in queries and retrieved passages.

7.1.4 Hybrid Retrieval Engine

- The system shall retrieve relevant passages from digitized classical texts using hybrid search combining:
  - Dense semantic search (FAISS vector database with cross-lingual embeddings)
  - Sparse keyword search (BM25 index for exact term matching)
- Implement medical relevance re-ranking to prioritize authentic classical passages.
- The system shall retrieve top-k passages (k=20) with score fusion and re-ranking to select top 5.
- Return passages with source metadata: text name, section, chapter number, verse range

7.1.5 Answer Generation Module

- The system will generate natural language answers using Retrieval-Augmented Generation (RAG).
- Prioritize extractive synthesis: direct extract and combine sentences from retrieved passages with minimal modification.
- Integrate retrieved context with user queries to generate coherent, faithful responses.
- Support multilingual output (Sinhala, English, code-mixed).
- The system shall include precise chapter and verse citations in standardized format for every claim.
- Display original Sanskrit text alongside translations (Devanagari, IAST, English, Sinhala)

7.1.6 Source Grounding and Explainability

- Provide structured citations: "Text Name, Section, Chapter Number, Verse Range" (e.g., "Charaka Samhita, Chikitsa Sthana, Chapter 3, verses 12-15") .
- Display original source passages in multiple scripts with translations.
- Implement faithfulness verification: ensure >85% of answer content is directly supported by retrieved passages.
- Generate confidence scores based on retrieval quality, source agreement, and faithfulness.
- The system shall detect and explicitly present contradictions when different texts disagree.
- Highlight entity mentions interactive glossary providing definitions and cross-references.
- Provide related passages and cross-references to other relevant chapters

7.1.7 Citation Verifications and Quality Assurance

- Verify citation accuracy: ensure chapter and verse numbers correctly reference claimed information.
- Map every answer sentence to specific source passages for transparency.
- Flag low-confidence answers (<0.6) or low-faithfulness answers (<0.8) for review.
- The system enables users to navigate from citations to full text passages.
- Provide visual indicators showing which parts of answer come from which sources.

7.1.8 Expert Validation and Feedback

- Provide interface for practitioners to validate answer accuracy and citation correctness.
- Flag low-confidence or low-faithfulness responses for expert review.
- The system shall integrate expert feedback into quality improvement cycles.
- Maintain annotation history and inter-rater reliability metrics for expert validation

7.2 Non-Functional requirements

- Scalability: Manage 10,000+ document pages and 100+ concurrent queries.
- Reliability: Effectively process multilingual and code-mixed queries with >90% uptime.
- Usability: Offer user-friendly, multilingual dashboard accessible to patients, students, and practitioners.
- Performance: Return responses within 2-3 seconds for typical queries.
- Security: Encrypt user data, anonymize query logs, ensure HIPAA-compliant handling of health information.
- Source Verifiability: Provide precise chapter/verse citations and original text display for every answer.
- Faithfulness: Maintain >85% faithfulness score (proportion of claims supported by sources).

- Citation Accuracy: Achieve >90% accuracy in chapter/verse references (expert-validated).
- Maintainability: Adopt modular architecture (Document Processing, Retrieval, Generation, Explainability) for debugging and upgrades.
- Cultural Validity: Ensure outputs align with authentic Ayurvedic principles and classical interpretations.

7.3 Use Case

Use Case ID	01
Use case name	Responding to User's Ayurvedic Inquiry with Source Citations
Preconditions	• Classical Ayurvedic corpus (Charaka Samhita, Sushruta Samhita, Ashtanga Hridaya, Madhava Nidana) has been processed through document intelligence pipeline. • Texts indexed with chapter/verse metadata and citation mappings. • NER, hybrid retrieval, and RAG models trained and integrated. • QA dashboard launched and accessible to users
Primary Actor	End User (patient, student, or practitioner)
Main Scenario	1. User logs into QA dashboard. 2. User submits health-related question in Sinhala, English, or code-mixed format (e.g., "Pitta වැඩි වුනාම කන්න ඕන foods මොනවද?" / "What foods balance Pitta?"). 3. System preprocesses query: language detection, code-switching resolution, term normalization, Sanskrit equivalent mapping. 4. System applies NER to identify key entities (e.g., Pitta dosha, dietary recommendations). 5. Hybrid retrieval engine searches corpus using dense semantic search + sparse keyword matching. 6. Top - 20 passages were retrieved, re-ranked by medical relevance, top - 5 selected . 7. RAG model synthesizes answer through extractive approach, combining sentences from retrieved passages. 8. Faithfulness verification checks that answer claims are supported by sources (target >85%) . 9. System generates structured citations for each claim: "Charaka Samhita, Sutra Sthana, Chapter 25, verses 40-45" . 10. System displays response with: - Natural language answer (2-4 paragraphs) - Source citations with chapter/verse references - Original Sanskrit passages (Devanagari + transliteration + translation) - Confidence score (High/Medium/Low) - Related passages for further reading. 11. User can click citations to view full passage context.

	12. If confidence <0.6 or faithfulness <0.8, system flags response for expert review. 13. Expert validates answer accuracy and citation correctness. 14. Validated responses stored; feedback integrated into quality metrics
Extensions	(2) System seamlessly handles Sinhala-English code-mixed queries through query enhancement. (6) If multiple relevant passages are found across different texts, system fuses evidence and cites all sources. (8) When texts disagree, system presents both perspectives with respective citations. (10) System detects if evidence is insufficient and explicitly states uncertainty. (12) Low-confidence queries flagged and prioritized for expert validation
Post conditions	• Users receive verifiable answers with precise source citations enabling independent verification. • Query and answer logged with faithfulness and citation accuracy metrics. • System improves through expert validation feedback. • Citation index updated if corrections needed

Table 4 Use Case

8 CONCLUSION

The proposed Source-Grounded Question Answering System for Classical Ayurvedic Literature bridges traditional Ayurvedic wisdom with modern computational techniques while maintaining scholarly rigor and verifiability. By incorporating document intelligence for classical manuscript processing, cross-lingual semantic search, and retrieval-augmented generation with faithfulness verification, the system delivers trustworthy, source-grounded, and multilingual responses.

Unlike generic QA systems or AI chatbots that may hallucinate information, this system prioritizes transparency and verifiability through precise chapter and verse citations, enabling users to trace every claim back to authentic classical texts. The hybrid RAG approach combines factual accuracy from retrieval with natural language fluency from generation, while faithfulness verification ensures >85% of content is directly supported by sources.

The system addresses critical gaps in existing Ayurvedic digital resources:

- First computational framework for processing multi-script classical manuscripts (Sanskrit Devanagari, Sinhala).
- Cross-lingual query support enabling Singlish/Sinhala queries on Sanskrit texts.
- Source-grounded answers with verifiable chapter/verse citations.
- Novel evaluation metrics: Citation Accuracy (>90%) and Faithfulness Score (>85%)

Through document intelligence, the system digitizes classical texts into machine-readable, citation-indexed formats, laying groundwork for computational Ayurveda research. Expert validation and iterative quality improvement ensure alignment with scholarly standards and clinical requirements.

This research contributes three novel components to the field:

1. Document Intelligence Pipeline for classical manuscript processing with layout preservation and citation indexing .
2. Cross-Lingual Semantic Search enabling colloquial queries on Sanskrit classical texts.
3. Source-Grounded Answer Generation with verifiable citations and faithfulness verification.

Ultimately, the project aims to make authentic Ayurvedic knowledge accessible, verifiable, and reliable in the digital era, supporting education, research, and evidence-based traditional medicine practice.

9 BUDGET AND BUDGET JUSTIFICATION

The table below depicts the overall budget of the entire proposed system

Component	Amount (USD)	Amount(LKR)
Data Preparation	200	61488.78
Computational Resources	250	76860.98
Cost of Internet	80	24595.51
Software and Tools	120	36893.27
Total	650	199838.54

Table 5 The expected budget analysis of the proposed system

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